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Modeling Renewable Energy Production for Base Stations Power Supply

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***Abstract*—Cellular access networks need to reduce their dependence on the grid, with the twofold objective to decrease operational cost and guarantee self-sustainability in case of grid unreliability. For doing so, an interesting possibility is to use renewable energy generators. The power supply system considered here consists of small units that power individual Base Stations (BSs) and are composed of solar renewable energy (RE) generators combined with energy storage units to sustain the BS operation during night or low production periods. The simulation and design of the power supply system require the availability of simple, yet accurate, models of the RE production. In this paper, we propose a stochastic model of the daily RE production that is represented by a single random variable, characterized by the mean value and the variance, that multiplies a shape function describing the production in the timeframe of a day. Our results show that the model, despite being simple, is accurate in predicting the system performance. Moreover, our simulations reveal that, while the mean value of production is fundamental to discriminate an insufficient-production from a sufficient-production regime, the variance of the production has an important impact on the system performance. Higher values of the variance imply a reduced BS self-sufficiency; in these cases, larger PV panels are required to guarantee a sufficient RE production regime.**

I. Introduction

Nowadays several research efforts are devoted to the deployment of effective solutions to make mobile networks more energy efficient and more sustainable. To this extent, the integration of RE in mobile networks to power Base Stations (BSs) results to be a promising approach, given that the BSs account for the largest fraction of energy consumption in mobile access networks, but they individually require relatively little power supply that can be provided by small generators. It is expected that the worldwide annual amount of new installations of RE power supply for BSs will increase up to around 84,000 in 2020, becoming more than 6 fold higher with respect to 2012 [1]. Mobile system implementations where BSs are powered by RE derived by the conversion of solar radiation, by means of photovoltaic (PV) panels, are currently adopted mainly in areas where the traditional power grid is either unreliable or not available, like in remote regions in emerging countries. Batteries are used to store extra amounts of harvested energy for future usage, such as during no production periods (like during night). The resiliency of similar systems to power outages and their capacity to reduce or nullify the energy demand from the grid can be evaluated only in relation to the RE production, which is location dependent. Indeed, the analysis leading to a proper system dimensioning, both in terms of PV panel and battery storage size, strictly depends on a reliable characterization of the RE production profile.

The growing interest in the integration of RE generators in mobile networks is confirmed by the increasing number of works related to this topic [2], [3]. Various studies are available in the literature specifically addressing the dimensioning of PV panels and storage capacity, in relation to the mobile system energy need, the possibility of taking energy from the traditional grid and the constraints imposed by costs. Real profiles of sun irradiance are often used to estimate the solar energy that can be harvested depending on the PV size and either simulate [4]–[6] or model [7] the behavior of the storage system [8].

It is known that RE production and PV system performance are highly affected by weather variability; for this reason, predicting and modeling this variability is a quite critical task. Many studies in the literature provide stochastic models of solar irradiance, rather than energy production, over different time scales. In [9] both the short-term and long-term variations of daily solar power are characterized. In [10], the authors start from data collected during twenty years in a region in Algeria and model the monthly production using several distributions: the Weibull, Gamma, Normal, logistic, lognormal and loglogistic distributions. Some of these distributions are used in [11] for predicting daily solar irradiance, for six different locations in Taiwan that have limited seasonal variability. While modeling a hybrid wind/solar system, in [12] a Beta distribution is used to model solar irradiance observed on a shorter time scale of the order of a few hours. Beta distribution for solar irradiance is used also in [13] and in [14]. For [13], the focus is on a stochastic model for dimensioning hybrid wind/solar systems; the model takes into account also demand variability and the solar irradiance is considered at peak demand. In [14], the Beta distribution is used to model the hourly global solar irradiance. Stochastic models of hourly solar irradiance are proposed also in [15], that uses several distributions, including Exponential, Weibull, Geometric, lognormal, Gamma. A shorter time scale of 5 minutes is emplyed in [16], in this case the Normal distribution is used; one-minute irradiance is considered in [17].

In this paper, focusing on the PV powering system of a BS that is part of a mobile network, we propose a simple model of the energy produced by PV panels of relative small size and we investigate the impact of the energy generation variability on the power supply system performance and its dimensioning.

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The main contributions of the paper are:

- Starting from the study of data about the solar irradiance in the city of Torino, we propose a simple stochastic model of the amount of energy that is produced by PV panels. The model is based on the identification of a shape function that represents the typical production profile of a day; the function is then scaled through a random variable that represents the production daily variations. The model can be used at different time scales, e.g., for representing the daily or the hourly production.
- By using the model to represent the daily production of a PV system providing power supply to a BS, we investigate the impact of the energy generation variability on the system and its dimensioning. Our results, derived for various dimensioning of PV panels and storage, show that the mean and variance of the random variable representing the daily generation are the key parameters needed to predict, with high accuracy, the system performance; performance is evaluated, mainly, in terms of battery depletion probability and energy demand from the grid.

The paper is organized as follows. In Sec. II, the scenario of interest is presented. The model of the RE production is presented in Sec. III. The simulation environment used for the derivation of the results in discussed in Sec. IV, while the results are shown in Sec. V. Finally, Sec. VI concludes the paper.

II. Scenario

The scenario considered in this study consists of BSs providing mobile coverage over a given area and powered by solar energy. Each BS is provided with a set of PV panels, capable of producing RE from sun irradiation to power the BS itself. Furthermore, each BS is equipped with storage capability (a set of batteries), to address the typically intermittent production of RE. The BSs have access to a power grid, so that the BS demand can be satisfied even in case the amount of produced RE alone is not sufficient for the normal BS operation. This work investigates a portion of the described scenario, made up by a single BS with its set of PV panels and batteries, although the results can be scaled up to an extended access network. We first deploy a simple stochastic model of the RE production obtained from PV panels, accounting for the real sun irradiation and its location based variability. This model is then used to simulate the behavior of the RE powered system and to evaluate its performance under different RE production profiles. A typical LTE BS is considered; since BSs power consumption profile has limited variability with traffic load [18], we assume a constant power demand of 1 kW. The BS is equipped with PV panels, made up by multiple modules enabling the conversion of solar radiation into electricity with an efficiency that depends on the adopted PV technology. The nominal capacity of a PV module corresponds to the maximum DC output power obtained by the PV device under standardized environmental conditions and it is commonly measured in peak Watt [Wp]. Considering some among the most efficient

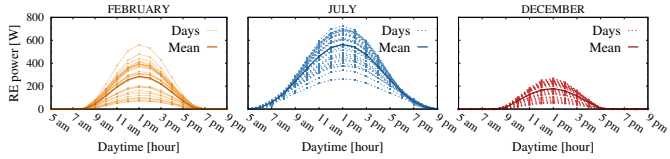

Fig. 1: Daily profiles of RE power production [W] per 1 kWp in different sample months in Torino (results obtained with PVWatts).

commercial PV modules built with traditional technologies (crystalline silicon), whose efficiency may achieve values as high as 20% [19], a PV panel surface of about 4.9 m^2 can be assumed per each kWp of PV panel capacity in locations like those in Italy that we consider in this paper [5].

The storage capacity is provided by lead-acid batteries, a storage technology widely adopted for PV systems [20]; storage elements with capacity 200 Ah and voltage 12 V are considered. Battery represents a rather critical component of the PV system, especially in relation to its charging efficiency and lifecycle duration. Indeed, the charging efficiency depends on the state of charge (SOC), showing an optimal operation point at a rather high SOC level (~80% [21]), which also prevents battery ageing [22] and capacity loss [20]. However, due to the naturally intermittent production of RE and the need to continuously power the BS, lead-acid batteries undergo frequent deep charging and discharging cycles, thus operating under partial SOC and inducing a variable charging efficiency over time. This behaviour also affects the battery service time, since the lifetime duration decreases inversely with the depth of discharge (DOD), dramatically shortening for DOD>80%. A tight constraint of maximum DOD≤50-70% allows to reach up to 1000 battery cycles of lifetime duration. An overall charge efficiency of 85% is commonly assumed for lead-acid batteries [21]; when considering also the discharging process, the total energy efficiency is estimated to be 75% [23], meaning that for each energy unit (1 kWh) drawn from the storage, around 1.33 kWh of RE must have been produced.

III. Modeling RE production

In this section, we propose a model to characterize RE production, with its variability. The model is derived from the analysis and understanding of data obtained from PVWatts [24] about RE production during the typical meteorological year (TMY) in Torino. While fitted and tested for Torino, the model can in general be used to represent the production in locations in which, similar to Torino, the intra-day weather variability is limited. As is shown below, the model consists of two elements: a shape function that represents the profile of production in a day, and a random variable that represents the daily variability and it is a scale factor to be applied to the shape function.

As mentioned above, the model is derived from data obtained by PVWatts, a tool that provides, on an hourly basis, the average levels of RE power production obtained from a standard PV panel with capacity of 1 kWp over the typical

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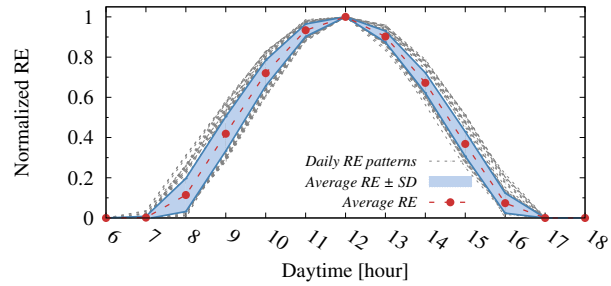
year. A crystalline silicon technology is assumed for the PV modules. A DC-to-AC inverter efficiency of 96% is considered. Given an optimal setting of tilting and azimuth angles for the considered location (20° and 180°, respectively), an average daily solar radiation of 4 kWh/m^2 is observed over the whole year. Typical performance losses found in real systems, about 14%, are taken into account.

Fig. 1 shows sample curves of the RE production during different days, curves are grouped by some months representative of different seasons, namely February, July and December. The solid line in each graph reports the average curve for the selected month. Each month has its own variability, but, clearly, seasonal variations are even more significant, with much higher production in summertime with respect to winter months. To deploy a model for a proper dimensioning of PV panels and batteries, a worst-case approach is taken and the season in the year with the lowest RE production is considered. For the selected location, such period is represented by the months of November, December and January (92 days), when an average solar radiation of 1.18 kWh/m^2 per day is achieved. Let us denote as RE_p the RE power production [W] at peak time, for a unitary PV panel (1 kWp) and denote with RE_d the daily total amount of RE production [kWh]. An average RE_p of about 193 W per 1 kWp is observed (range [42,349] W); as a reference consider that the mean value for the whole year is 390 W (range [41,760] W).

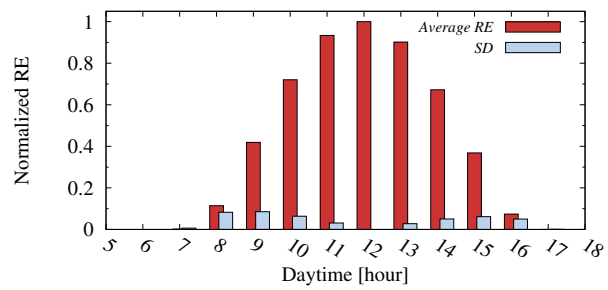
To investigate the similarity of the daily patterns of the RE production during the 3-month period, Fig. 2a reports the daily curves of RE production, with sample granularity of 1 hour, normalized with respect to the RE_p value of each day; the curve of the mean normalized RE production for the whole period is reported along with the range corresponding to *mean value* $\pm SD$. Given the RE_p, the *shape* of RE production is very similar in each day. The production in a given day can then be represented by the red dashed line (obtained by averaging the production hour by hour) multiplied by a factor that corresponds to the RE_p of that day. Hence, RE_p is the only parameter needed to represent the daily production and to distinguish the production in different days. Fig. 2b reports also the standard deviation of the normalized RE production, for each hour of the day. SD is always $< 10\%$ of RE_p, and it is larger as we get closer to the curve tails, with a maximum of 8.6% of the normalized RE_p. These variations are negligible with respect to the central peak hours, in terms of total daily RE production. The curve representing the average normalized RE production during the 3-month period, denoted by $f_N(t)$ (red dashed curve in Fig. 2a), can be derived by averaging the normalized production for each time in a day and interpolating the new obtained points. The actual RE production curve, denoted as $f(t)$, represents the real RE pattern production in a specific day:

$$f(t) = RE_p \cdot f_N(t). \tag{1}$$

The *support* of the RE production curves is related to the duration of the daylight time; in our case, its average is about

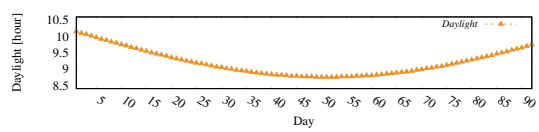


(a) Daily patterns of normalized RE production

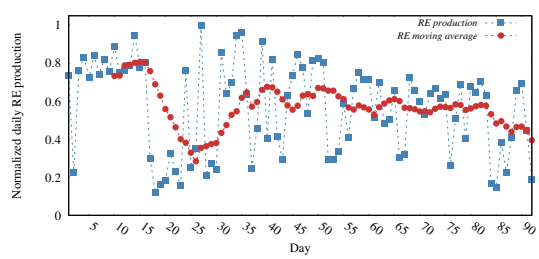


(b) Hourly normalized RE production

Fig. 2: Daily normalized RE production patterns (average and standard deviation) in the cold season.



(a) Daylight duration.



(b) Normalized daily RE production.

Fig. 3: Daylight time duration across the three months and normalized daily RE production with its moving average.

9 hours. The selected 3-month period is characterized by a period of daylight ranging from 8 h and 45 min to 10 h and 9 min. A maximum variation of just ±10% with respect to the mean value (9 h and 12 min) is observed. Such a relatively low variability range does not seem to have a direct and significant impact on the RE production variability, as it can be seen from Fig. 3. The blue curve in Fig. 3b represents the value of the daily production, RE_d, normalized with respect to the maximum RE_d in this period (that is 1.76 kWh), while the red curve is the corresponding moving average over the previous 10 days; the orange curve in Fig. 3a corresponds to the duration of daylight for each day. The moving average

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varies within a pretty limited range, hence resulting rather unrelated to the daylight time variation, which, by converse, is first decreasing and then increasing in the same period. The lowest values of the moving average are actually seen around November 27th and January 31st, in two moments in which the daylight time is longer than in those days showing higher RE production. Therefore, the small differences in the daylight duration among days in this period do not affect significantly the level of daily production in the corresponding days. These observations allow us to assume a profile with a constant support over the considered period. Given the fixed common shape for the normalized daily RE production (holding for the whole period), $f_N(t)$, the actual daily RE production pattern can be represented as in (1) by multiplying $f_N(t)$ by a random variable that represents RE_p. Denoting the r.v. by A, a sample a_i of A defines the amplitude of RE_p in day i and the production curve in day i is given by $a_i f_N(t)$. The r.v. A is simply defined by the probability distribution function. We will show in what follows that if we simply fit the mean and the variance of the r.v. A to those of the peak production RE_p, we obtain a simple model that well captures the system behavior.

Summarizing, the proposed model for the RE production consists in obtaining a daily profile of production by multiplying the shape function $f_N(t)$ by a r.v. A whose mean and variance are set equal to those typical of the peak production in the considered location. The proposed model can be applied for locations with limited intra-day weather variability, like in the case of Torino and several other locations, according to PVWatts traces.

IV. Simulating the system

To prove the utility of the model, we now apply it to a case of practical interest, a BS equipped with PV panels and battery storage. We analyze the system behavior to understand how the RE production profile affects its performance. In this section, we describe the simulation environment used for deriving the results.

To define the system we use the following parameters: C_{BS}, the BS power consumption (in kW), assumed to be constant; S_{PV}, the PV panel size (in kWp); B, the storage size (in kWh), intended as the portion of the whole capacity provided by the batteries that can actually be exploited. Indeed, the storage amount defined by this parameter is limited by the maximum Depth of Discharge (DOD) allowed for each battery element and it may range from 50% to 70% of nominal capacity, depending on the minimum battery life duration that we are willing to accept. The RE production is characterized by using the model described in the previous section, keeping into account that variable amounts of RE are produced each day, depending on the panel size, S_{PV}, and on the peak production distribution in the considered location, RE_p. Furthermore, the proposed model assumes that each day the RE is produced only during a limited period of time (*daytime*), whereas during the rest of the day (*nighttime*) no RE can be produced. The amount of RE that is produced during daytime, denoted as R, is derived as $R = RE_d \cdot S_{PV}$. According to the proposed model, RE_d can be related to RE_p by means of a constant coefficient, F:

$$RE_d = F \cdot RE_p = \int_D f_N(t) \cdot RE_p$$

where D is the support of $f_N(t)$, i.e. the daytime period, varying with latitude. $f_N(t)$ represents the curve of the normalized RE production per kWp in a given location. In this work, F is derived in the worst-case period for the city of Torino, where latitude is 45.2°N and $daytime$ is about 9 h. The locally produced RE can be used to directly power the BS; the extra amount is stored in the battery to be used later on, if there is still space available. In case of battery depletion, the BS demand can be satisfied by taking energy from the grid. Storage charging and discharging losses are taken into account, considering a battery efficiency of 85%. In our simulation, the storage level, along with other performance parameters, is sampled daily, both at the beginning of daytime (the most critical point for RE availability), denoted by $Sd_{(i)}$ for day i, and at the beginning of nighttime, $Sn_{(i)}$. The battery levels are related by:

$$Sd_{(i)} = min(B, max(0, Sn_{(i-1)} - Cn_{(i-1)}))$$

where $Cn_{(i-1)} = C_{BS} \cdot nighttime$ is the BS energy consumption during nighttime. The storage level at the beginning of nighttime, is computed for day i as:

$$Sn_{(i)} = min(B, max(0, Sd_{(i)} + R_{(i)} - Cd_{(i)}))$$

where $Sd_{(i)}$ is the storage level at the beginning of daytime in the same day, $R_{(i)}$ is the RE produced during daytime and $Cd_{(i)} = C_{BS} \cdot daytime$ is the daytime BS energy consumption. For BS service continuity and reasonably low values of the depletion probability, the storage size B must be large enough to store at least the amount of energy needed to cover the nighttime demand, when no RE is produced.

V. Performance analysis

In this section, we analyze the performance of the system, focusing on the impact of the main stochastic characteristics (mean and variance) of the daily energy production.

Several simulations were run under many different configuration settings of input parameters, and considering different types of distribution for the random variable A that represents the daily production RE_p. In particular, to test the impact of the distribution of energy production, normal (ND) and uniform distributions (UD) have been tested, under multiple combinations of the values of mean and variance. The BS consumption is assumed to be $C_{BS} = 1$ kW. Results from these two distributions have been compared with those derived from the simulation under the empirical distribution (ED) obtained from real traces of sun radiation in Torino. 10,000 samples were collected on a daily basis for each simulation, both at the beginning of daytime and of nighttime, but only the first kind of results are reported, since they correspond to the worst-case. The system performance has been evaluated by means of the following performance indicators: B_O, average daily battery

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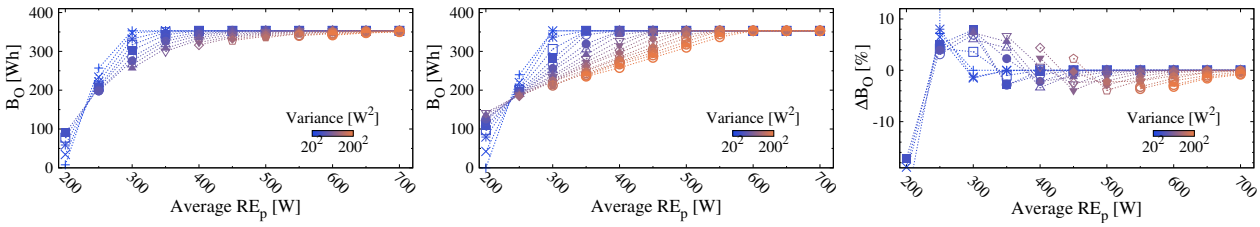

(a) ND of RE_p. (b) UD of RE_p. (c) Δ_{B_O} between ND and UD case.

Fig. 4: Battery occupancy, B_O, under normal (ND) and uniform distribution (UD) of RE_p and Δ_{B_O} versus average RE_p in scenario 1.

occupancy; Q_{90}, 90th percentile of battery occupancy; P_D, storage depletion probability; G, average daily grid demand. Two scenarios, 1 and 2, have been selected to represent the general trend of the results. In scenario 1, the storage capacity is B = 18 kWh, and the panel size is S_{PV} = 24 kWp; in scenario 2, B = 20 kWh, S_{PV} = 26 kWp.

a) Transition from scarce to sufficient RE production regime: Fig. 4 shows the average battery occupancy (B_O) versus the mean daily production in scenario 1, both in case of normal distributed production and uniformly distributed production. Each curve represents a different variance. Clearly, the average charge B_O increases with the average production, with a much faster increase for lower values of the mean and of the variance. In particular, for low mean production, a higher variance allows higher levels of charge, whereas, starting from a transition point in the RE_p (between 200 W and 250 W), there is an inversion such that, for higher variance, lower charge levels are observed. The reason is that when production is low, variance helps in giving more chance to fill the battery; conversely, when production is enough, variance causes more chances not to fill the battery.

The same considerations hold under the other distributions. However, although the general trend is similar, for low values of the variance of the production, v_R, the ascent is slightly steeper under UD rather than ND. Similar trends are observed in scenario 2 (data not reported for lack of space), featuring larger PV panel and storage size. To better evaluate the differences between the two types of RE_p distribution, Fig. 4c reports the relative difference observed in the values of B_O under ND versus UD, denoted as Δ_{B_O}, computed for each configuration of $[RE_p, v_R]$ in scenario 1. Except for the lowest value of production (200 W), Δ_{B_O} is never larger than ±10%, a quite small value. This shows that the main impact of the variability of the production can be quite well represented simply by the mean and the variance of the daily production.

Results about depletion probability, P_D, are reported versus the average RE_p in Fig. 5, in scenario 2 and under UD of RE_p. The trend is the opposite with respect to the behavior of the battery occupancy B_O, since depletion probability decreases for increasing values of the mean production, RE_p, with a huge decrease and steep descent for low values of average RE_p. Again, the transition point in RE_p can be identified between 200 W and 250 W, below which the depletion

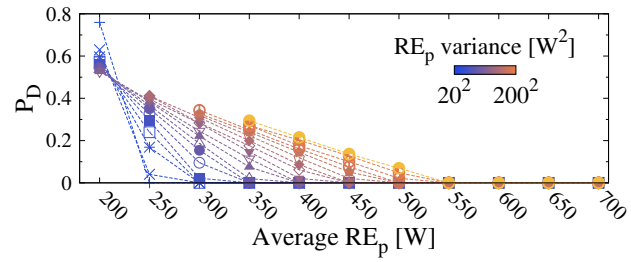

Fig. 5: Depletion probability, P_D, under UD of RE_p in scenario 2.

probability remains smaller than 50% for any value of the variance, v_R. Results show that larger values of the battery and PV panel (scenario 2) allow to more rapidly reduce depletion probability. As observed before, the general trend is similar under ND and UD, but in the UD case for low values of the variance of production, the descent is steeper than for ND, while it becomes more gradual, with respect to the ND case, for higher values of the variance v_R. The relative difference of P_D between ND and UD case, Δ_{P_D} results to be higher than for B_O, but still within ±30%; higher values of Δ_{P_D} are found only when P_D is low (less than 6%).

The identified transition point corresponds to the switching between two regimes: a RE production not sufficient to significantly satisfy the BS need and a regime providing a sufficient amount of energy for BS operation.

b) Effects of RE_p variance: The system is now investigated in terms of how its performance is affected by the variance of production v_R, for fixed average RE_p. Fig. 6a shows the trend of B_O versus the variance for scenario 2 in case of uniform distribution of RE_p. Each curve represents a different value of average RE_p. It can be observed that for high average values, B_O tends to gradually decrease as v_R grows larger. For intermediate values of RE_p, the descent becomes faster, especially for small v_R. For the lowest v_R values, B_O tends to rapidly increase, instead of decreasing, as v_R becomes higher. The opposite trend is seen for P_D (data not shown). P_D rapidly decreases as v_R increases. In general, relatively low values of depletion probability (smaller than 15%) can be achieved only for average RE_p higher than a minimum threshold (~250 W in the presented results, but it would be higher in case of smaller S_{PV}). However, v_R may deeply affect

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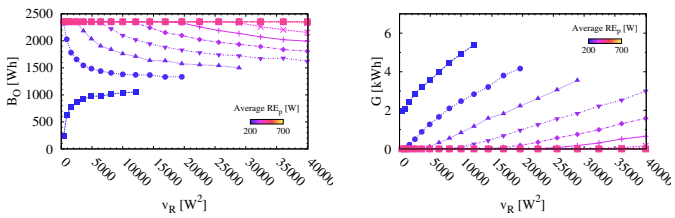
the P_D that can be observed. This means that, for a given value of average production, as v_R increases it becomes more and more difficult to have the BS off-grid, unless high values of mean RE_p are made available, for example by increasing the PV panel size. A very similar behaviour can be observed for B_O and P_D in case of normally distributed average production.



(a) B_O, UD (b) G, UD

Fig. 6: Battery occupancy, B_O, and grid demand, G, under uniform distribution (UD) of RE_p.

The average daily grid demand, denoted as G, is shown in Fig. 6b in case of UD. G increases as v_R becomes larger, with sharper ascent for low RE_p values. The curve corresponding to RE_p=200 W clearly shows that, for low RE_p, in no case the produced RE is enough to make the BS completely off-grid, unless by increasing the PV size: in fact, even with the lowest v_R, at least ~2kWh must be required from the grid. Notably, despite the difference in the P_D curve slopes between ND and UD case, the G curves result to be much more overlapped even under the two different distributions (ND data not reported), meaning that although the frequency of energy requests from the grid may slightly differ in the two cases, the actual average amount of energy taken from the grid is approximately the same. Wider differences in B_O between ND and UD are seen for higher v_R. This results more evident from Fig. 7, showing Q_{90} versus v_R for various RE_p mean values and comparing ND (curves with circle points) versus UD (curves with star points) case. The transition between a constant not null Q_{90}~2.3kWh (which is the maximum possible level of residual storage remaining after the nighttime BS consumption) and Q_{90}=0kWh is very sharp, especially when v_R is low with respect to the mean RE_p. The v_R corresponding to this transition point tends to become higher for larger mean RE_p, until the average RE_p reaches values such that Q_{90} never becomes null. Furthermore, for the same mean RE_p, the switching point becomes higher and higher under ND with respect to UD case, as v_R grows larger. This behaviour is consistent with results presented in Fig. 6, where P_D appears to increase faster with raising v_R under UD rather than ND case, that seems slightly outperforming UD to this extent. However, considering the similar trends shown for G under ND and UN, the difference in Q_{90} between ND and UD case may not be relevant, especially for small RE_p, unless an off-grid target is pursued.

c) Testing the system under empirical distribution of RE_p: Finally, to further validate the model and confirm the

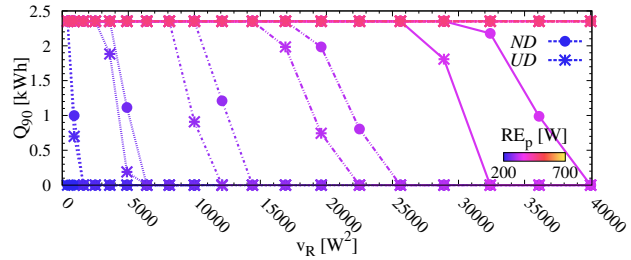


Fig. 7: 90th percentile of battery occupancy, Q_{90}, under normal (ND) and uniform distribution (UD) of RE_p in the scenario 2 (B=20 kWh, PV=26 kWp).

previous findings, we test the system behavior under the empirical distribution (ED) of RE_p, as it is derived from real data of RE production in Torino in the worst-case 3-month period. Fig. 8 shows the probability density function (pdf) and cumulative distribution function (cdf) of RE_p computed from the empirical data. Two peaks can be identified in

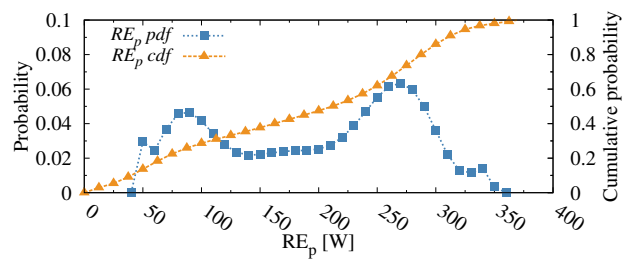
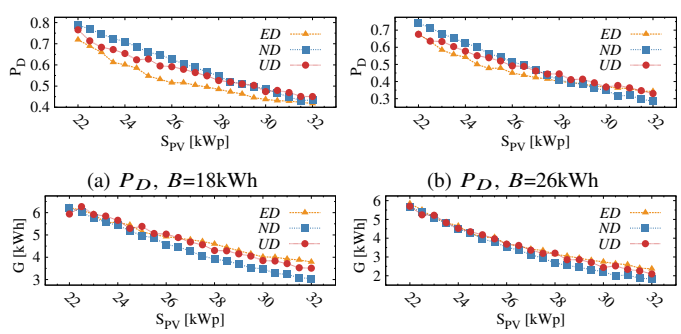


Fig. 8: Peak production RE_p: pdf and cdf of the empirical distribution



(a) P_D, B=18kWh (b) P_D, B=26kWh

(c) G, B=18kWh (d) G, B=26kWh

Fig. 9: Depletion probability, P_D, and grid demand, G, under ED, ND and UD of RE_p, for B=18 kWh and B=26 kWh.

the pdf, likely corresponding to the prevalent types of daily weather conditions during the whole period. We now compare the results obtained with the ED of energy production against the other two considered distributions, ND and UD, under the same values of mean and variance of the daily production. In Fig. 9, P_D and G versus S_{PV} are reported for the three types of distribution and two battery sizes (B=18 kWh and B=26 kWh). The patterns are very similar and curves almost overlap.

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These results show that the proposed model, despite being simple, is accurate and representative of the RE production in a location, like Torino, in which the daily weather is quite stable. As indicated in (1), the model is simply given by two components: a function that gives the shape of the production in a day, and a random variable that, multiplied by this function, gives the actual value of the production. In addition, most of the system behavior can be caught by simply focusing on the mean and variance of this random variable.

VI. Conclusions

The integration of RE into power supply systems of cellular access networks is needed for cost reduction and for improving the energy self-sufficiency of mobile access networks. However, to facilitate the system design and a proper PV panel and storage dimensioning, simple and reliable RE production models are needed. In this paper, we proposed a simple stochastic model where RE production is represented by a shape function multiplied by a random variable, characterized by a mean value and a variance. This random variable characteristics depend on the sun irradiation of the considered location. Our results show that the model is representative of RE production in locations with low intra-day weather variability. Indeed, when considering different distributions of daily production with the same mean and variance, very limited differences can be observed.

Simulations revealed also the relevance of RE production variability: for fixed mean production, higher values of the variance imply a reduced BS self-sufficiency and larger PV panels are hence required to move to a sufficient RE production regime.

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